Transmission Through a One–Dimensional T-Stub and its Analytic Structure

1 Introduction

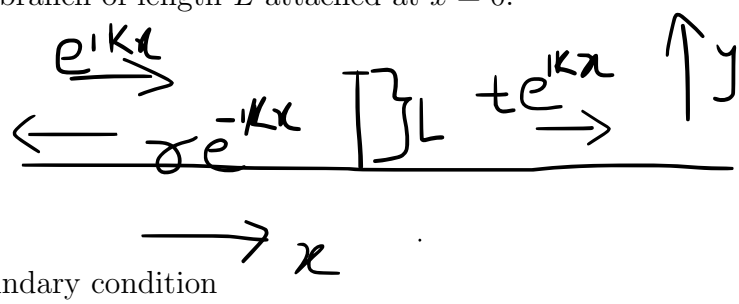
A simple but instructive quantum scattering system is a one–dimensional quantum wire with a side–attached stub. The geometry forms a "T-junction" and is therefore known as a T-stub problem.

The system is important because it provides a minimal model in which interference between a continuum channel and a discrete resonant state produces an asymmetric line shape known as a Fano resonance.

2 Geometry of the Problem

Consider a one–dimensional wire with a side branch of length L attached at $x = 0$.

The three regions are



- Left lead: $x < 0$
- Right lead: $x > 0$
- Stub: $0 < y < L$

The stub terminates with a hard-wall boundary condition

$$\psi(y = L) = 0.$$

3 Wavefunctions

The stationary Schrödinger equation gives plane wave solutions.

3.1 Left lead

$$\psi_L(x) = e^{ikx} + re^{-ikx}$$

3.2 Right lead

$$\psi_R(x) = te^{ikx}$$

3.3 Stub

To satisfy $\psi(L) = 0$ we choose

$$\psi_S(y) = A \sin[k(L - y)].$$

4 Boundary Conditions

At the junction the wavefunction is continuous

$$\psi_L(0) = \psi_R(0) = \psi_S(0).$$

Thus

$$1 + r = t = A \sin(kL).$$

Current conservation at the junction requires

$$\partial_x \psi_L(0) - \partial_x \psi_R(0) + \partial_y \psi_S(0) = 0.$$

Substituting derivatives gives

$$ik(1 - r) - ikt - Ak \cos(kL) = 0.$$

Using $A = t/\sin(kL)$ we obtain

$$ik(1 - r) - ikt - kt \cot(kL) = 0.$$

Using $1 + r = t$ yields the transmission amplitude

$$t(k) = \frac{2i}{2i + \cot(kL)}.$$

5 Transmission Probability

The transmission probability is

$$T(k) = |t(k)|^2.$$

This gives

$$T(k) = \frac{4 \sin^2(kL)}{4 \sin^2(kL) + \cos^2(kL)}.$$

6 Transmission Zeros

Transmission vanishes when

$$\sin(kL) = 0.$$

Thus

$$k_n = \frac{n\pi}{L}.$$

These energies correspond to standing waves inside the stub. At these energies destructive interference blocks transmission completely.

7 Analytic Structure of the Transmission Amplitude

The transmission amplitude

$$t(k) = \frac{2i}{2i + \cot(kL)}$$

is analytic in the complex k plane except for poles where

$$2i + \cot(kL) = 0.$$

Thus the scattering amplitude contains

- zeros from $\sin(kL) = 0$
- poles from $2i + \cot(kL) = 0$

The pole–zero structure is responsible for the asymmetric resonance.

8 Connection to Fano Resonance

Near a resonance energy E_r the transmission amplitude takes the universal form

$$t(E) \approx \frac{E - E_0}{E - E_r + i\Gamma/2}.$$

The transmission probability then becomes

$$T(E) = \frac{(q + \epsilon)^2}{1 + \epsilon^2}$$

where

$$\epsilon = \frac{E - E_r}{\Gamma/2}.$$

This asymmetric profile is the characteristic Fano resonance.
Physically this arises from interference between

- direct transmission through the wire
- resonant scattering through the stub

9 Exercise: Plotting the Resonance Line Shape

Plot the transmission probability

$$T(k) = \frac{4\sin^2(kL)}{4\sin^2(kL) + \cos^2(kL)}$$

as a function of kL.
Observe the following:

- Transmission zeros at $kL = n\pi$
- Resonant peaks between the zeros
- The asymmetric resonance profile characteristic of Fano interference

MATLAB Code

```
clear
L = 1;

kL = linspace(0,6*pi,2000);

T = (4*sin(kL).^2)./(4*sin(kL).^2 + cos(kL).^2);

plot(kL,T,'LineWidth',2)

xlabel('kL')
ylabel('Transmission T')
title('Transmission through a T-stub')

grid on
```

10 Discussion

The plot shows sharp transmission zeros and resonant peaks. These features arise from interference between the direct transmission path and the resonant state in the stub.

Thus the T-stub provides a simple physical realization of Fano resonance in quantum transport.